USN. | | | | | | | | | | [18MTEEM202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE–2019
PHOTOVOLTAIC AND SOLAR THERMAL ENERGY

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i. Answer FIVE full Questions. Question No. 1 and 2 are compulsory.**
ii. Missing data may be suitably assumed.

1. a. Define the following with equations: **10**

i. Slope
ii. Altitude angle
iii. Zenith angle
iv. Solar azimuth angle
v. Surface azimuth angle
vi. Declination angle
vii. Hour angle
viii. Solar altitude angle
ix. Slope
x. Solar constant

b. Calculate the values of Ig, Ib and Id for a horizontal surface on a clear day between 09.00-10.00 hrs at Nagpur (21°06´N, 79°03´E). on June 20th, 1968, given A=1092W/m^2; B=0.185 W/m^2; C=0.137 W/m^2 **10**

2. a. Explain the following with respect to Solar cells: **10**

i. P-N junction potential.
ii. Width of the depletion region.
iii. Band to band recombination.
iv. Trap assisted recombination.

b. Distinguish between the following: **10**

i. Direct band gap and indirect band gap semiconductors.
ii. Intrinsic semiconductors and extrinsic semiconductors.

3. a. Explain briefly the different processes involved in the fabrication of Si solar cells. **10**

b. Explain with a neat sketch the production of monocrystalline Si ingots through C_Z process. **10**

OR

4. a. Explain with a neat schematic sketch the production of Electronic Grade Silicon (EGS) using Siemen's process. **10**

b. Briefly explain with neat sketches any two methods of producing Si sheets. **10**

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5. a. Discuss the concept of mismatch occurring in PV modules when connected in series and in parallel. **12**

b. Explain the effect of solar irradiation and temperature on the power output of solar PV modules. **08**

OR

6. Explain any four important components of Balance of systems (BoS) required for Solar PV installations. **20**

7. a. With neat sketches explain any four types of concentrating collectors. **10**

b. With neat sketches, explain direct and indirect gain forced circulation solar dryers. **10**

OR

8. a. Explain with a neat sketch, the working principle of forced circulation water heating system in closed loop configuration. **10**

b. Explain with a neat sketch, the working principle of solar industrial process heating system using a concentrating collector array. **10**